

prior to component placement. The solder paste is printed through a thin stencil to achieve the required pattern. Although it seems like a simple process, this is the area where errors and delays are likely to happen. Hence it is important to keep tab of the latest technologies and choose the right one for SMT production.

High-speed, automatic printers with 12-second cycle time and 10-micron repeatable accuracies at fine pitch represent the current technology, according to Bhatnagar.

Specialised high- and low-volume printers are available as well as models that can handle both. Models such as Panasonic's SMT screen printers include high-speed, multi-snap-off and support snap-off to give users the flexibility to realise both SMT screen printing and superfine pitch printing. Customisability is an important feature of current-generation printers.

Quick changeover, support for lead-free solder, selectable optimised stencil release feature, automated cleaning, 2D and 3D inspection are some of the recent features.

Six-channel reflow

Reflow ovens are used for heating and soldering of PCBs by application of controlled heat and cooling. The PCB or substrate is transported through a series of temperature-controlled zones. This gradually increases the temperature of the PCB and solder through a programed temperature profile to achieve optimum solder paste melting and joint formation.

Today highly precise temperature profiling and data collection for reflow soldering applications is possible using four- or six-channel thermal profilers. These can be used to accurately record temperatures at multiple locations on the surface of PC boards or specific components via multiple type-K thermocouples connected from the assembly to a shielded, temperature-resistant data-logging unit. This data is extremely important to control the processes and helps avoid faults.

A friend in software

"Software makes the difference in modern SMT solutions. The software determines how flexible your SMT solution is and how efficiently it supports the processes. The SIPLACE software suite, for example, puts the industry's most wide-ranging and powerful software library at your fingertips for significant productivity improvements," comments Chua. It includes program packages for product definition, optimisation and line control, production monitoring and process control, setup verification and traceability, order and material management, etc.

Quick and easy software integration is a must these days. Open interfaces are also essential to ensure that the machinery and the software can be integrated with other systems.

Assembléon machine has built-in PPM calculator. These machines can also be integrated with third-party ERP packages directly and thus user can get online report of material requirement.

"Control systems and management data collection are important and complete in all aspects in all the products we offer," says Suresh Nair of Leptech Corporation, a Mumbai-based maker of SMT equipment.

On the buyer front, Bhatnagar comments on what level of software integration they expect in the equipment they buy: "Software integration requires more user interactive software. The equipment should have compatible interfaces to handshake with other standard equipment, universal data bus and data transfers, Ethernet interfaces and also enhanced statistical data capture capabilities. Integrated bar code traceability can give a smooth tracking of detection and counter-measure functions to help in achieving zero defects."

Despite most equipment coming with extensive software interfaces, there is still a need for standardisation. The Frost & Sullivan report mentions that "...there remains a need for collaborative efforts among SMT manu-

facturers to frame a unified interface standard, wherein machines can communicate more effectively with each other, ensuring seamless interoperability and flexibility."

Soft corner for green

Today, environment-friendly and energy-efficient SMT equipment are in great demand. Complying with directives such as the RoHS and Waste Electrical and Electronic Equipment Directive (WEEE) is more than a choice today. There is a variety of equipment favouring lead-free production. This includes soldering equipment, printers, SMT filters, etc. For example, SMT filters that survive higher solder reflow temperature profiles (SN95) so that their system PCBs would be RoHS compliant, are also available to comply with the needs of the electronics assembly and EMS companies.

The adjective 'green' involves several factors including energy-efficiency to conserve environmental resources. Energy efficiency, reduced carbon footprint, etc also have a very favourable micro benefit—reduced power bills for the company!

Way to go

It is clear that the SMT equipment space is on a high-run when it comes to technological development. What it requires to grow despite the economic downturn is to rethink and strengthen the business models. Some feel the need to go beyond selling SMT equipment and provide support services. Others suggest the exploration of relatively unaffected markets.

The real challenge now for the Indian industry, according to Bhatnagar, is to, "create a strong supply chain and maintain short turn-around times, especially by efficiently managing its operations with the overseas component base or a nascent distribution network in India." ●

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